

Ahmed Ibrahim

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EXPERIENCE

Weill Cornell Medicine – Qatar

Oct. 2023 – Present

AI Research Specialist

Doha, Qatar

- Processed multimodal datasets including text and 2D/3D medical images to enhance model development pipelines
- Developed and optimized language models for healthcare applications through pre-training and task-specific fine-tuning
- Designed and implemented a Retrieval-Augmented Generation (RAG) system using synthetic and real text data
- Built and evaluated Large Multimodal Models supporting visual question answering and automated report generation
- Architected AI agents enabling dynamic multi-step reasoning and advanced Agentic RAG workflows
- Contributed to research lifecycle including experimentation, analysis, and scientific paper preparation

Qatar University

Oct. 2023 – Dec. 2023

Research Assistant, Department of Mathematics, Statistics and Physics

Doha, Qatar

- Trained deep learning models for stroke lesion segmentation in MRI scans to improve diagnostic accuracy
- Contributed to a research publication exploring novel medical image segmentation methodologies
- Conducted literature review, methodology development, and experimental result analysis

Qatar University

Jun. 2023 – Oct. 2023

Research Assistant, Department of Computer Science and Engineering

Doha, Qatar

- Engineered experimental setups and simulations using MATLAB and Unity for immersive testing environments
- Developed prototypes of haptic devices integrating multiple feedback mechanisms
- Performed extensive literature reviews on Virtual Reality, Augmented Reality, and haptic feedback systems

PROJECTS

D3: A Small Language Model for Drug-Drug Interaction Prediction | Python, PyTorch, Pretraining, SFT 2025

- Pretrained and finetuned a compact ~70M-parameter Small Language Model for Drug-Drug Interaction (DDI) prediction, trained and evaluated on biomedical interaction data
- Benchmarked D3 against billion-parameter LLMs, showing the small specialized model achieves comparable accuracy at a fraction of the inference cost; published in *Machine Learning with Applications* (Elsevier, 2025)

MERA: Medical Electronic Records Assistant | Python, LangChain, RAG 2025

- Designed an end-to-end Retrieval-Augmented Generation assistant for clinical EHRs, unifying medical question answering, report summarization, and patient-case similarity search behind a single domain-specific retrieval pipeline
- Generated a synthetic clinical training set with Mistral, Qwen, and Llama, and validated on de-identified MIMIC-IV-Note records using correctness, groundedness, ROUGE-1 F1, Jaccard, and METEOR; published in *Machine Learning and Knowledge Extraction* (MDPI, 2025)

EDUCATION

Qatar University

Jan. 2019 – Jun. 2023

BSc in Computer Engineering

Doha, Qatar

- **Coursework:** Machine Learning, Object Oriented Programming, Data Structures, Robotics

SKILLS

Programming Languages: Python, R, C++

Frameworks & Libraries: PyTorch, TensorFlow, Hugging Face, LangChain, LlamaIndex

AI & Machine Learning: LLM Pre-training and Fine-tuning, Retrieval-Augmented Generation (RAG), Multimodal Models (VQA, Report Generation), AI Agents

Specializations: Segmentation (UNet, ViT), Natural Language Processing, Computer Vision, Medical Imaging

Data: Data Preprocessing and Curation, Data Visualization

AWARDS

Hackathon Winner: Health Hackathon

2nd Place: Senior Design Project Contest

2nd Place: Capstone Design Competition